

Element B - Documentation and Analysis of Prior Design Attempts

Project Title: MJ's Last Stand (Echo Rack)

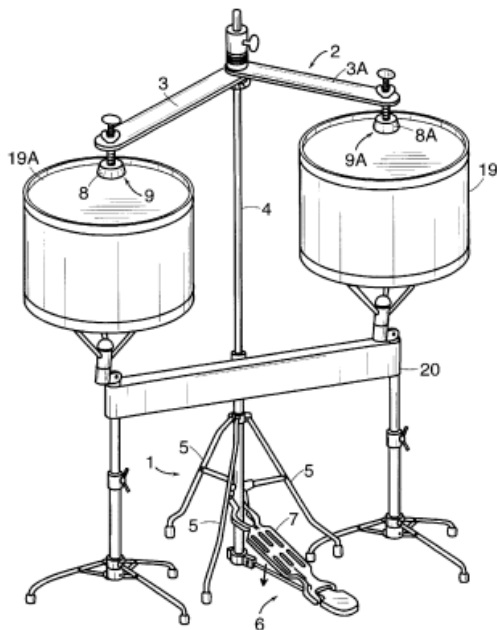
Prior Design Attempt #1 - Pedal Actuated Percussion Damper

Patent Number: 5986197

Source: Apa Format

Welcome To Zscaler Directory Authentication. (2025). Uspto.gov.

<https://ppubs.uspto.gov/dirsearch-public/print/downloadBasicPdf/5986197?requestToken=eyJzdWIiOiJiNGM4YWZjMy1jY2ZjLTRlMmEtOTVlOS04MTZhZjM5Zjc3ZDMiLCJ2ZXliOiJkMjQ3ZGU4My0xMjk2LTQ4NmEtYjVhZi05ODk5ZTgwNWY1YzYiLCJleHAiOiJB9>



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This patented design is a “pedal actuated percussion damper” that is made to pair with bongo, conga, and other sorts of drums. As the name suggests and as shown within the given diagram, with the press of a pedal, two damper arms usually resting right above the drums are lowered into the heads of the drums. A spring force is used to keep the apparatus in its ‘resting’ position. The damper apparatus as described by the patent, is highly customizable and adjustable to suit the preferences of the user(s) and accommodate the musician’s needs. For example, in consideration of the musician’s height, the apparatus can be lowered or raised. But the most notable aspect of the design is the ability to adjust the damping effect of the apparatus through numerous ways. The material of the damping pieces (that actually make contact) can be wood, rubber, cork, yarn, etc. The shape of the damping pieces can be flat or round like a sphere. Wing nuts are used to allow for manual adjustment of the distance between the damping pieces and the arms on which they are held.

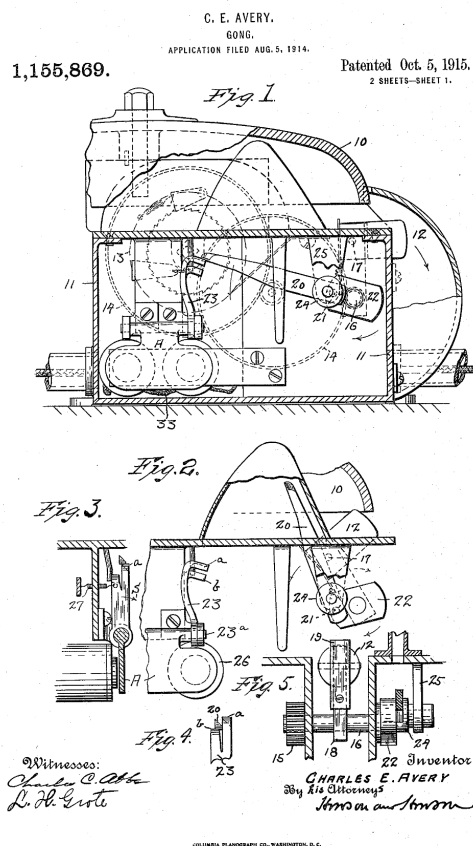
Although this design of a percussion mute is not exactly for a gong, it does provide a useful example for how a mute could be designed for a gong. This design isn't exactly a 'failure,' as I believe it is actually a nice and well thought out idea, but it simply does not fit our design criteria. We are not designing a mute for drums, but a gong, which are completely unique percussion instruments in nature and how they are played. The face of the gong does not face upward but sideways, meaning a pedal actuated mute/damper for a gong would require a translation of vertical motion to horizontal motion for the mute arms (which would likely have to be on pivots); we will require a unique design.

Prior Design Attempt #2 - Automatic Gong Striking & Muting Mechanism

Source: Apa Format

US1155869A - Gong. - Google Patents. (1914, August 5). Google.com.

<https://patents.google.com/patent/US1155869A/en>



Although it may seem confusing at first, this is a gong striking/muting mechanism patented in 1915 for commercial use. The design is very complex due to the small footprint, causing many things to be crammed into one space. Parts of the design oftentimes overlapped with each other and eventually caused failure. Specific repairs of the design were difficult to do

because you couldn't really reach the parts you were looking for without risking breaking it. The muting mechanism worked really well and it seems to just be two soft plates pushing on the gong from both sides to stop the ringing. The striking mechanism still worked, but was less useful due to the lack of options when it came to hitting the gong. (Things such as volume depend on where you hit the gong and how hard you hit it. There were options that the design did not have.)

This design wasn't exactly what we were looking for, however, it does provide insight on methods to mute the gong effectively. Muting the gong is a big part of our design and will be the most complex part of it. A simple stand isn't difficult to make, so although it's the most important part of the gong stand, it isn't what we are most worried about. We likely won't use most of the mechanical parts that this design had due to it being specifically designed for such a small space, but the way it mutes is very important and we will likely use that in our final design.

Prior design attempt #3 - Gel percussion Damper

Source: Apa Format

US5637819A - Percussion instrument damping - Google Patents. (1992, June 23). Google.com.
<https://patents.google.com/patent/US5637819A/en>:

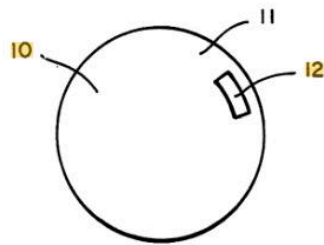


FIG. 1



FIG. 2

This design is a gel that dampens the sound creating by different percussion instruments. The gel dampens the vibrations the different instruments made creating less sound. When the gel is attached to the percussion instruments it compresses the instruments. The gel mostly consists of PVC copolymer resin. The causes the instruments to vibrate less, producing less sound. The gels come in different sizes. The size of the gel depends on the size of the percussion instrument. Smaller instruments would use a smaller amount of gel and larger instruments would use a large amount of gel. This helps maximize the sound dampening of the instrument. An advantage of this design is when the instrument is hit hard the gel will still adhere to the surface to the instrument. The gel turns the snare drum into a practice pad because it mutes the sound produced by the snare drum. This is very similar to a practice pad because it mutes the sound of a snare drum on a percussion allowing percussionist to practice without making a lot of noise.

Although this is a good design for battery percussion instruments such as the snare drum or bass drum this design would not work on a gong. This design does not match our criteria. We would like a muter or dapaner that is controlled with a pedal. Removing and Xy applying a large piece of gell would not be ideal when playing the gong. Although this design does not work for our gong stand it allows us to see different ways to mute percussion instruments. We could try to implement gel into our pedal depaning design for our gong stand.

Prior Design Attempt #4 - Modular self-dampening triggering system for acoustic percussion elements

Source: Apa Format

US10032440B2 - Modular self-dampening triggering system for acoustic percussion elements - Google Patents. (2016, May 2). Google.com.

[https://patents.google.com/patent/US10032440B2/en?q=\(Percussion+stand\)&oq=Percussion+stand](https://patents.google.com/patent/US10032440B2/en?q=(Percussion+stand)&oq=Percussion+stand)

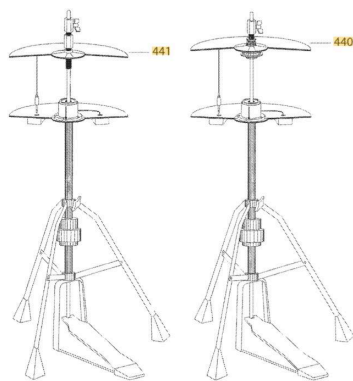
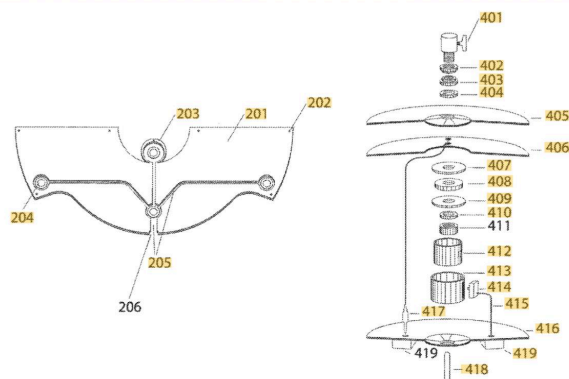


Fig. 4F

Prior Attempt



The prior design attempted to create an electronic cymbal that mimics the performance of an acoustic cymbal while allowing for volume control. The system includes:

- A metal cover plate that attaches to a cymbal or percussive element.
- At least one piezo sensor affixed to the cover plate.
- A dampening element (a flexible rubber sheet) positioned between the cymbal and cover plate, aligned with the piezo sensors.
- The goal was to maintain the natural feel of an acoustic cymbal while providing electronic triggering capabilities.

Several issues prevented this design from fully solving the problem of realistic cymbal emulation:

Striking Sensation – The prior design used hard resin electronic cymbals, which lacked the same tactile feel as traditional acoustic cymbals. The striking sensation was notably different, reducing realism.

Accuracy of Strike Detection – Detecting a strike with high accuracy was difficult due to limitations in the sensor placement and material choice. The design struggled to properly distinguish between different striking forces and positions.

Visual and Structural Issues – The combination of disk-shaped frames made the cymbal visually unappealing and possibly structurally less efficient than an acoustic design.

Performance Quality – The lack of an effective solution for mimicking dynamic responses in acoustic cymbals meant the electronic system could not fully replicate the expressiveness and nuanced playability of an acoustic cymbal.

For the gong stand, this prior design attempt suggests key considerations:

Dampening and Isolation: A flexible dampening element could be used to reduce unwanted vibrations while allowing for controlled resonance.

Sensor Placement: Proper alignment of piezo sensors is crucial for accurate strike detection, ensuring realistic electronic sound reproduction.

Material Selection: Unlike the prior attempt's hard resin, a hybrid of metal and flexible materials could improve realism and responsiveness.

Structural Aesthetics: Avoiding bulky or visually unappealing designs while ensuring durability is important for a functional and aesthetically pleasing gong stand.